Algebra 1: Polynomials - Practice Test

1. Simplify the expression $x^2 + 3x + 2$ by factoring.

A) $x(x+1)$
B) $x(x+2)$
C) $(x+1)(x+2)$
D) $(x-1)(x+2)$
E) $(x+1)(x-2)$

2. If $p(x) = x^2 - 4x + 3$, then $p(2) =$

A) -1
B) 1
C) 3
D) 5
E) 7

3. The polynomial $x^3 - 2x^2 - 5x + 6$ can be factored as

A) $(x-1)(x-2)(x-3)$
B) $(x-1)(x+2)(x-3)$
C) $(x+1)(x-2)(x-3)$
D) $(x-1)(x+1)(x-6)$
E) $(x+1)(x-2)(x+3)$

4. What is the value of x for which $2x^2 + 5x - 3 = 0$?

A) $\frac{1}{2}$
B) $-\frac{1}{2}$
C) $\frac{3}{2}$
D) $-\frac{3}{2}$
E) $\frac{1}{3}$

5. The equation $x^2 - 7x + 12 = 0$ has solutions $x =$

A) $3, 4$
B) $2, 5$
C) $2, 6$
D) $3, 5$
E) $4, 6$

6. The polynomial $x^4 - 2x^3 - 13x^2 + 14x + 12$ can be factored as

A) $(x-1)(x+1)(x-2)(x+6)$
B) $(x-1)(x-2)(x+2)(x+3)$
C) $(x-1)(x+1)(x+2)(x-6)$
D) $(x+1)(x-2)(x-3)(x+2)$
E) $(x+1)(x+2)(x+3)(x-2)$

7. What is the remainder when $x^3 + 2x^2 - 7x - 12$ is divided by $x + 3$?

A) 0
B) 1
C) 2
D) 3
E) -1

8. If $f(x) = x^3 - 2x^2 - 5x + 6$, then $f(-2) =$

A) -20
B) -16
C) -12
D) -8
E) -4

9. The polynomial $x^3 - 2x^2 - 11x + 12$ can be factored as

A) $(x-1)(x-2)(x+6)$
B) $(x-1)(x+2)(x-6)$
C) $(x-1)(x+1)(x-12)$
D) $(x+1)(x+2)(x+3)$
E) $(x-1)(x-4)(x+3)$

Open Ended Questions (FRQ)

FRQ 1. Prove that the polynomial $x^2 + 4x + 4$ can be written in the form $(x + a)^2$ for some real number a. Provide the value of a and simplify the given polynomial.

Hoca Notu / İpucu